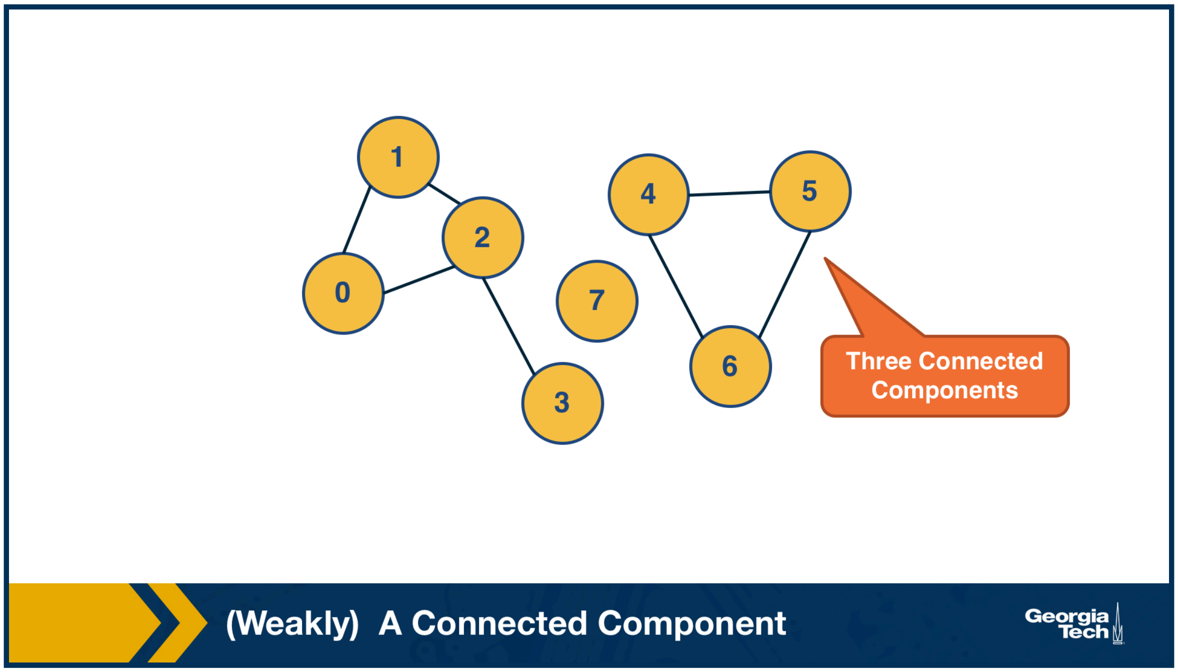
L2: (Weakly) Connected Components

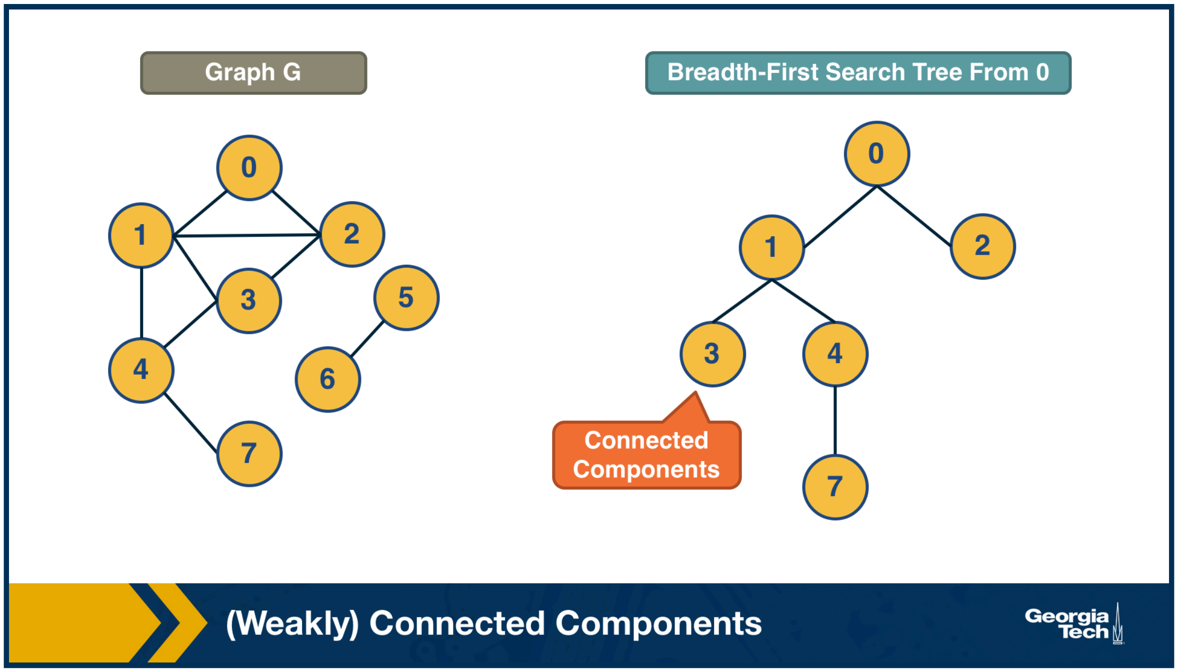
An undirected graph is **connected** if there is a path from any node to any other node. We say that a directed graph is **weakly connected** if and only if the graph is connected when the direction of the edge between nodes is ignored. It follows that if a directed graph is strongly connected, it is also weakly connected. In undirected graphs, we can simply refer to connected components.

As we saw in Lesson-1, there are many real-world networks that are not connected – instead, they consist of more than one ***connected components***.



A **breadth-first-search** *(BFS)* traversal from a node s can be used to find the set of nodes in the connected component that includes s. Starting from any other node in that component would result in the same connected component.

If we want to compute the set of all connected components of a graph, we can repeat this BFS process starting each time from a node s that does not belong to any previously discovered connected component. The running-time complexity of this algorithm is **Θ**(m+n) time because this is also the running-time of BFS if we represent the graph with an adjacency list.



Food-for-thought: If you are not familiar with the O, Ω, Θ notation, please read about them at: https://en.wikipedia.org/wiki/Big_O_notation (https://en.wikipedia.org/wiki/Big_O_notation)